

CS3329: Midterm Exam, HUST Spring 2026 Section A

Maximum Score: 30

Time: 45 Minutes

Name:

Troy ID:

Write clearly. Staple together and submit this question paper with the solutions.

1. [7 points] Describe an algorithm that, given n integers in the range 0 to k , preprocesses its input and then answers any query about how many of the n integers fall into a range $[a, b]$ in $O(1)$ time. Your algorithm should use $\Theta(n + k)$ preprocessing time.

2. [4 points] Solve using the Master's theorem:

$$T(n) = 2T\left(\frac{n}{3}\right) + n \log n$$

Master Theorem:

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

for some $\epsilon > 0$, $k \geq 0$, and $c < 1$,

- (a) If $f(n) = O(n^{\log_b a - \epsilon})$, then $T(n) = \Theta(n^{\log_b a})$.
- (b) If $f(n) = \Theta(n^{\log_b a} \log^k n)$, then $T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$.
- (c) If $f(n) = \Omega(n^{\log_b a + \epsilon})$, and $af\left(\frac{n}{b}\right) \leq cf(n)$, then $T(n) = \Theta(f(n))$.

3. [2 + 5 points] Consider a modification to merge sort in which n/k sublists of length k are sorted using insertion sort and then merged using the standard merging mechanism, where k is a fixed/constant value.

- (a) Show that the n/k sublists, each of length k , can be sorted by insertion sort in $\Theta(nk)$ worst-case time.
- (b) Show that the sublists can be merged in $\Theta(n \log(n/k))$ worst-case time.

4. [6 points] By following the divide-and-conquer approach, devise an algorithm to find the minimum value in an array of elements. Then also analyze this algorithm by writing a recurrence relation and solving it.

5. [6 points] Suppose that the splits at every level of quicksort are in the proportion $1 - \sigma$ to σ where $0 < \sigma < 1/2$ is a constant. What is the minimum and maximum depth of a leaf in the recursion tree?

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1. [4 points] Solve the following recurrence using the **Master Theorem**:

$$T(n) = 4T(n/2) + n$$

Identify a , b , and $f(n)$, state the correct case, and give the final asymptotic bound.

Master Theorem:

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

for some $\epsilon > 0$, $k \geq 0$, and $c < 1$,

- (a) If $f(n) = O(n^{\log_b a - \epsilon})$, then $T(n) = \Theta(n^{\log_b a})$.
- (b) If $f(n) = \Theta(n^{\log_b a} \log^k n)$, then $T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$.
- (c) If $f(n) = \Omega(n^{\log_b a + \epsilon})$, and $af\left(\frac{n}{b}\right) \leq cf(n)$, then $T(n) = \Theta(f(n))$.

2. [5 points] Consider **Quick Sort**.

Construct an input where Quick Sort runs in $\Theta(n^2)$ and explain why.

3. [6 points] Suppose that you have a black-box worst-case linear-time median subroutine. Give a simple, linear-time algorithm that solves the selection problem for an arbitrary order statistic.
4. [8 points] Consider the **KMP pattern matching algorithm**.

- Compute the LPS (Longest Prefix Suffix) array for the pattern:

AAACAAAA

- Explain how the LPS array helps avoid redundant comparisons.
- Briefly justify why KMP runs in $O(n + m)$ time.

5. [7 points] Consider modifying the PARTITION procedure by randomly picking three elements from array A and partitioning about their median (the middle value of the three elements).

Approximate the probability of getting at worst an α -to- $(1 - \alpha)$ split, as a function of α in the range $0 < \alpha < 1$.